Georgia Tech College of Engineering

COE 3001

MECHANICS OF DEFORMABLE BODIES

Lecture 10 **–** Torsion – Statically Indeterminate Problems

Yazhuo Liu

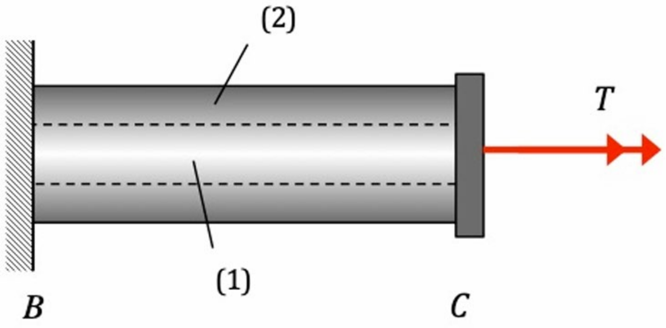
Georgia Institute of Technology

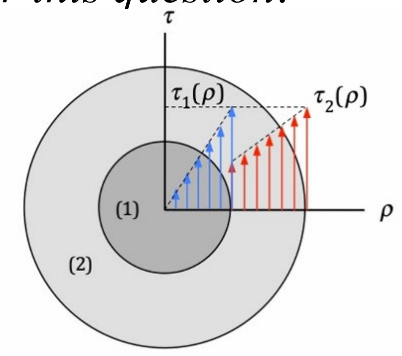
Feb. 18, 2026

Review

A shaft is made of a shell (2) and core (1) with an axial torque T applied at the connector C. Let G_1 and G_2 represent the shear modulus of (1) and (2), respectively. **There is no relative sliding between shell (2) and core (1).** The shear stress distribution across the shaft cross-section is shown in the figure. Note from the figure that the maximum shear stresses in (1) and (2) are equal. Choose the correct answer below:

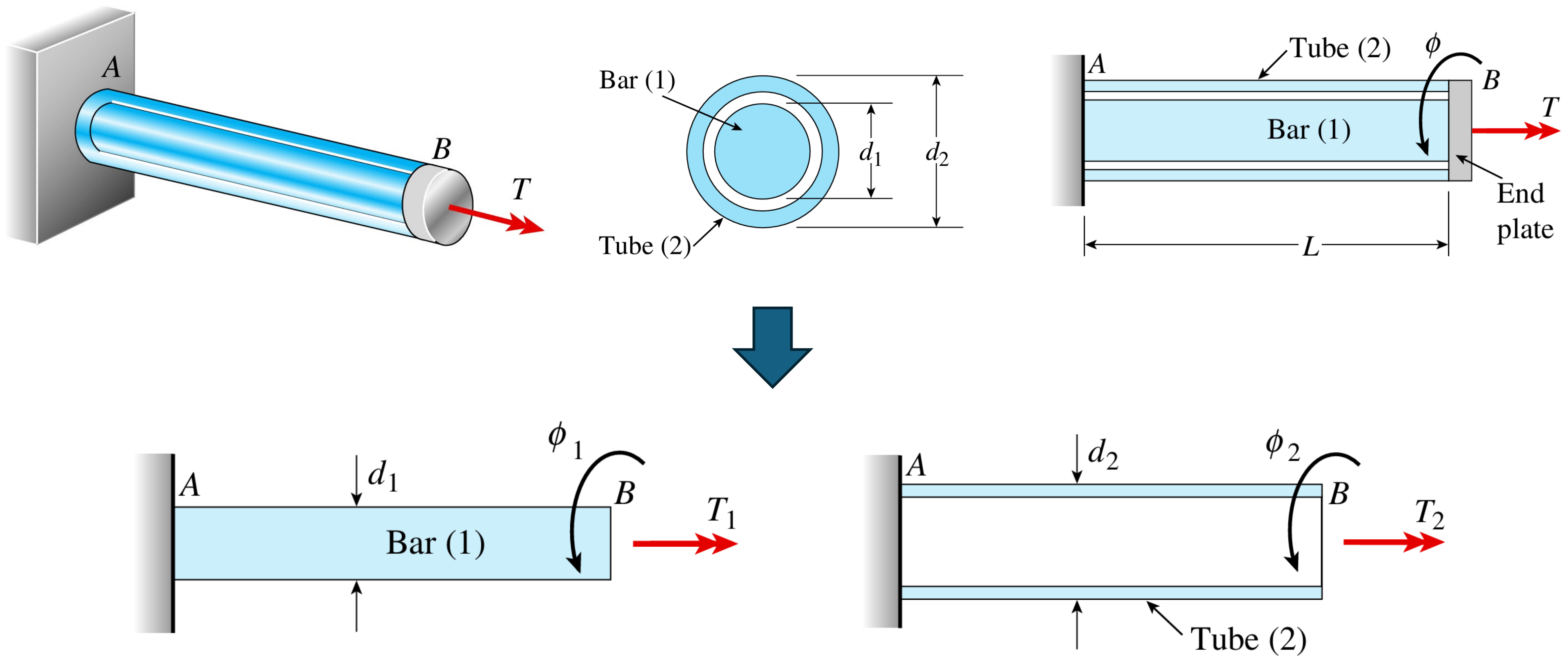
a) $G_1 > G_2$
b) $G_1 = G_2$
c) $G_1 < G_2$
d) *More information is needed in order to answer this question.*



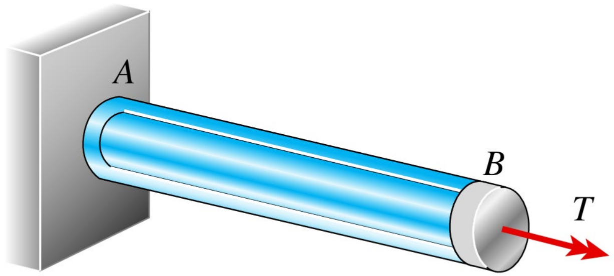


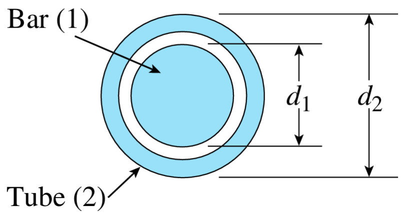
shear stress distribution on the shaft cross-section

Statically indeterminate bar in torsion



Standard procedure





① Draw FBD

② Write down all static equilibrium equation(s)

③ Calculate *degree of static indeterminacy*

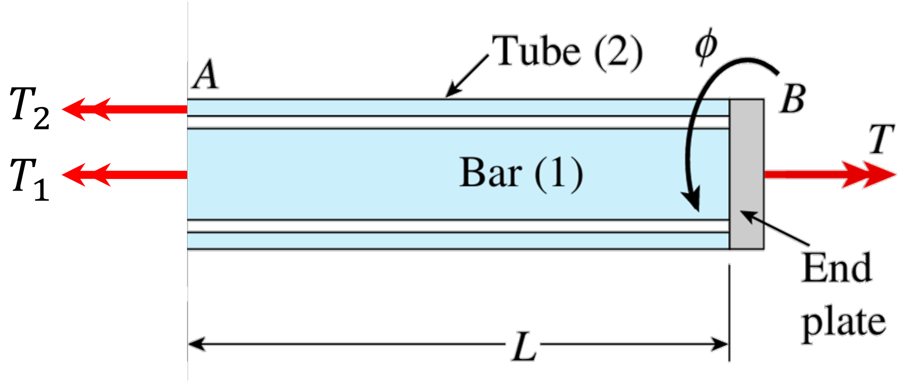
④ Find out the compatibility equation(s)

⑤ Using Force-displacement relation, convert compatibility equation(s) into equations of reaction force(s)

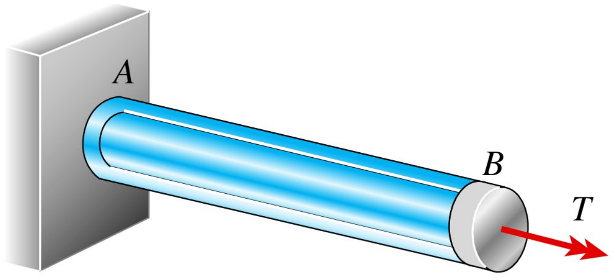
⑥ Solve all reaction force(s)

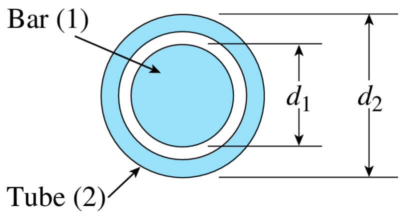
⑦ Calculate all required stress and strains

① FBD:



Standard procedure





① Draw FBD

② Write down all static equilibrium equation(s)

③ Calculate *degree of static indeterminacy*

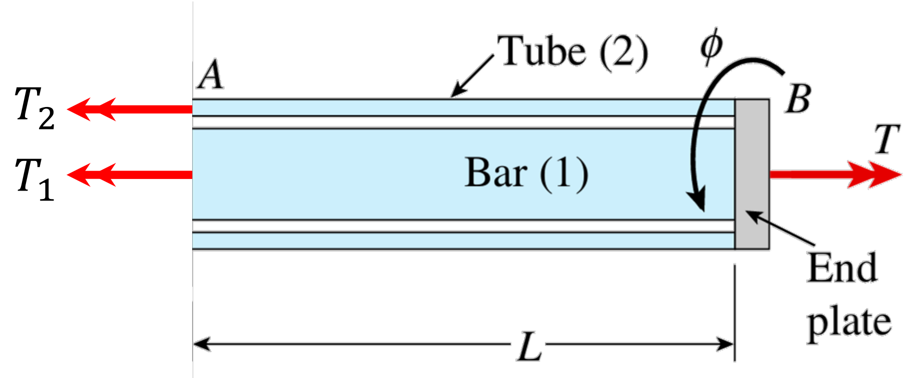
④ Find out the compatibility equation(s)

⑤ Using Force-displacement relation, convert compatibility equation(s) into equations of reaction force(s)

⑥ Solve all reaction force(s)

⑦ Calculate all required stress and strains

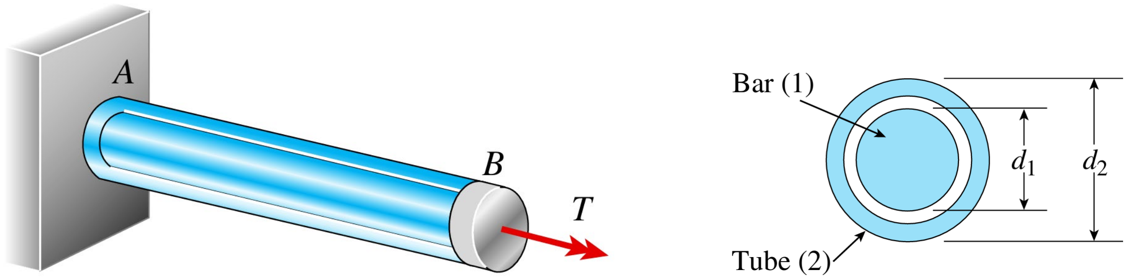
① FBD:



② Equilibrium equation: $T_1 + T_2 = T$

③ Unknowns: T_1 and T_2, *degree of static indeterminacy* = 1

Standard procedure



① Draw FBD

② Write down all static equilibrium equation(s)

③ Calculate *degree of static indeterminacy*

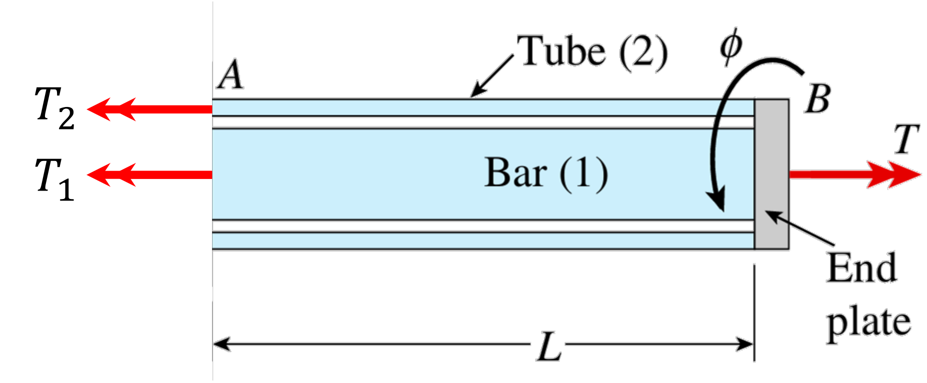
④ Find out the compatibility equation(s)

⑤ Using Force-displacement relation, convert compatibility equation(s) into equations of reaction force(s)

⑥ Solve all reaction force(s)

⑦ Calculate all required stress and strains

① FBD:

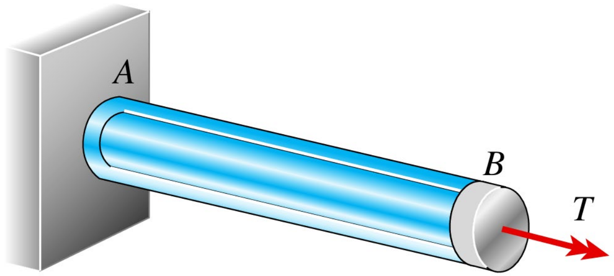


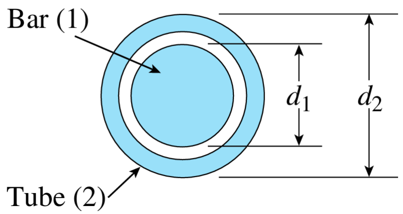
② Equilibrium equation: $T_1 + T_2 = T$

③ Unknowns: T_1 and T_2, *degree of static indeterminacy* = 1

④ Compatibility equation: $\phi_1 = \phi_2$

Standard procedure





① Draw FBD

② Write down all static equilibrium equation(s)

③ Calculate *degree of static indeterminacy*

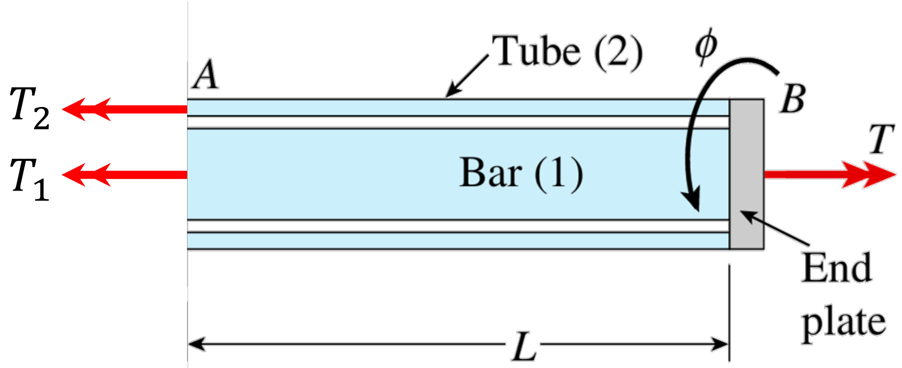
④ Find out the compatibility equation(s)

⑤ Using Force-displacement relation, convert compatibility equation(s) into equations of reaction force(s)

⑥ Solve all reaction force(s)

⑦ Calculate all required stress and strains

① FBD:



② Equilibrium equation: $T_1 + T_2 = T$

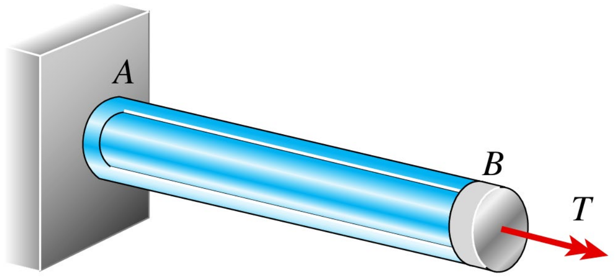
③ Unknowns: T_1 and T_2, *degree of static indeterminacy* = 1

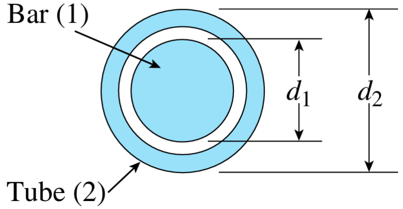
④ Compatibility equation: $\phi_1 = \phi_2$

⑤ Using torque-displacement relation:

$$\phi_1 = \frac{T_1 L}{G I_{p1}}, \qquad \phi_2 = \frac{T_2 L}{G I_{p2}}, \qquad \Longrightarrow \frac{T_1 L}{G I_{p1}} = \frac{T_2 L}{G I_{p2}}$$

Standard procedure





① Draw FBD

② Write down all static equilibrium equation(s)

③ Calculate *degree of static indeterminacy*

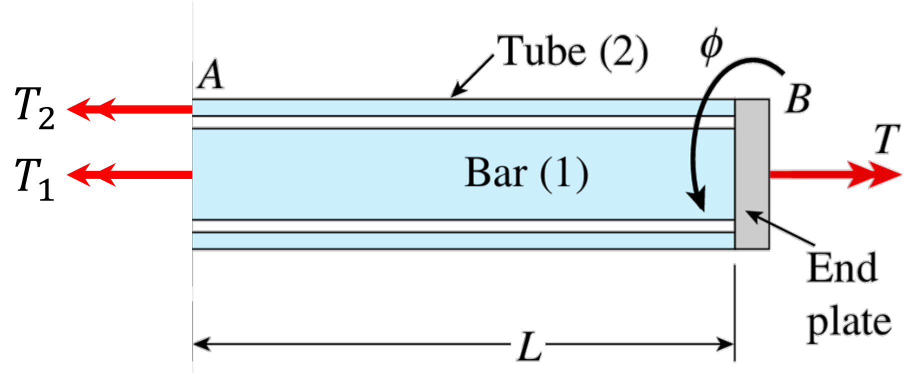
④ Find out the compatibility equation(s)

⑤ Using Force-displacement relation, convert compatibility equation(s) into equations of reaction force(s)

⑥ Solve all reaction force(s)

⑦ Calculate all required stress and strains

① FBD:



② Equilibrium equation: $T_1 + T_2 = T$

③ Unknowns: T_1 and T_2, *degree of static indeterminacy* = 1

④ Compatibility equation: $\phi_1 = \phi_2$

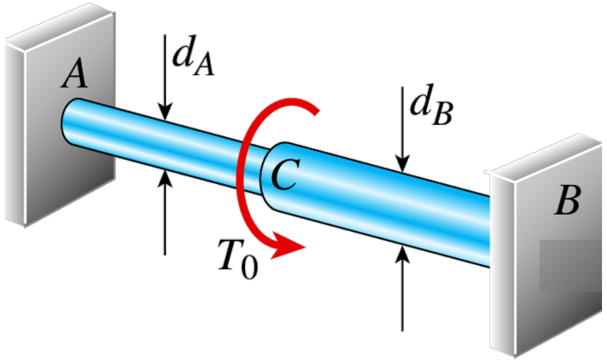
⑤ Using torque-displacement relation:

$$\phi_1 = \frac{T_1 L}{G I_{p1}}, \qquad \phi_2 = \frac{T_2 L}{G I_{p2}}, \qquad \Longrightarrow \frac{T_1 L}{G I_{p1}} = \frac{T_2 L}{G I_{p2}}$$

⑥ Solve T_1 and T_2

$$T_1 = T\left(\frac{G_1 I_{p1}}{G_1 I_{p1} + G_2 I_{p2}}\right) \qquad T_2 = T\left(\frac{G_2 I_{p2}}{G_1 I_{p1} + G_2 I_{p2}}\right)$$

Exercise

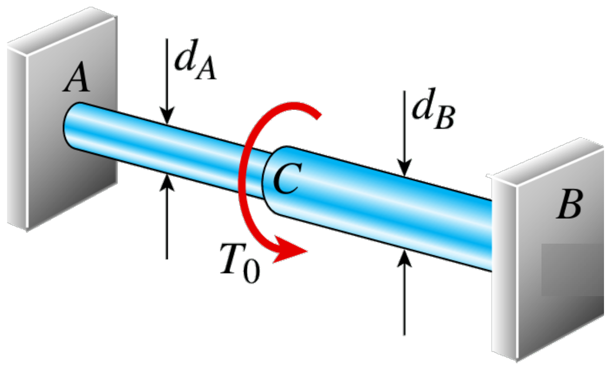


Given: T_0, d_A, d_B, L_A, L_B, I_{pA}, I_{pB}, G

Solve:

- T_A and T_B
- TMD and TDD

Exercise

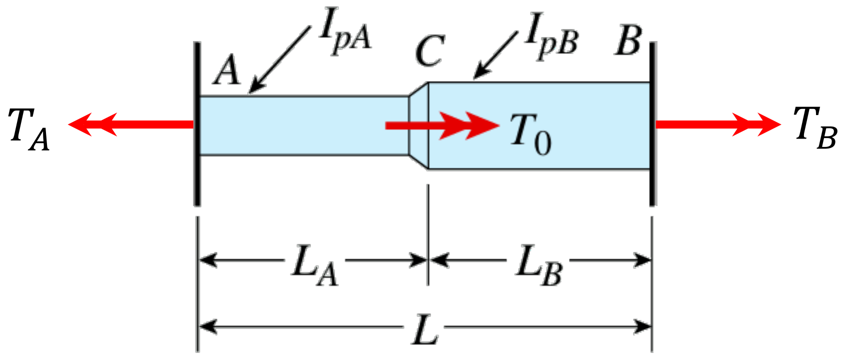


Given: $T_0, d_A, d_B, L_A, L_B, I_{pA}, I_{pB}, G$

Solve:

- T_A and T_B
- TMD and TDD

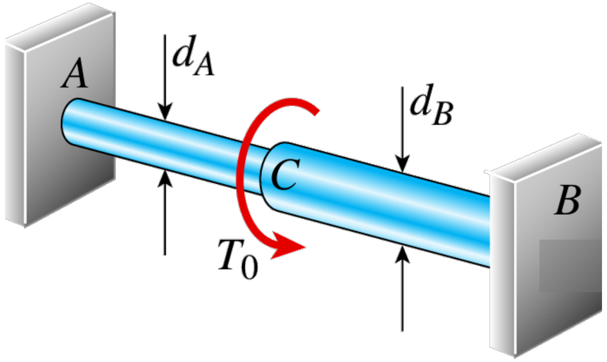
① FBD:



② Equilibrium equation: $-T_A + T_0 + T_B = 0$

③ Unknowns: T_A and T_B, *degree of static indeterminacy* = 1

Exercise

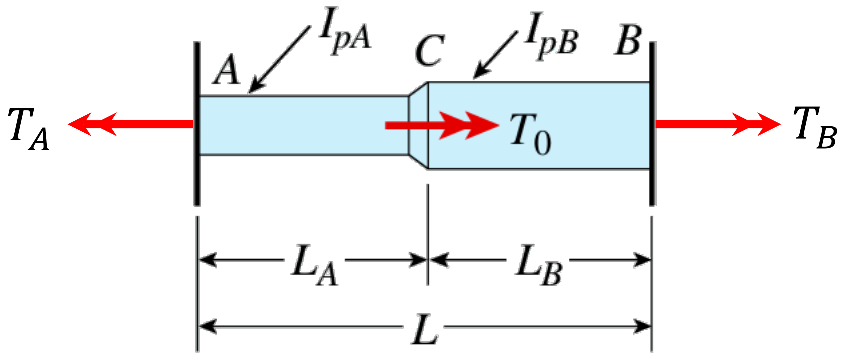


Given: T_0, d_A, d_B, L_A, L_B, I_{pA}, I_{pB}, G

Solve:

- T_A and T_B
- TMD and TDD

① FBD:

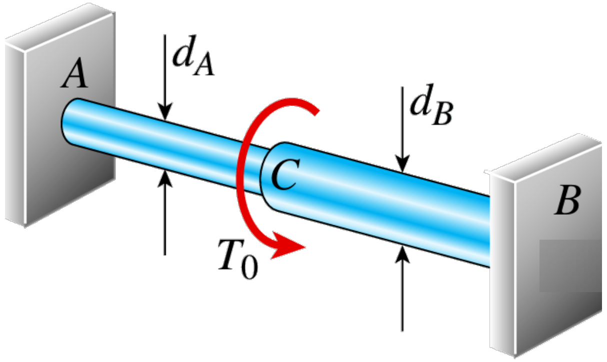


② Equilibrium equation: $-T_A + T_0 + T_B = 0$

③ Unknowns: T_A and T_B, *degree of static indeterminacy* = 1

④ Compatibility equation: $\phi_{AB} = 0$

Exercise

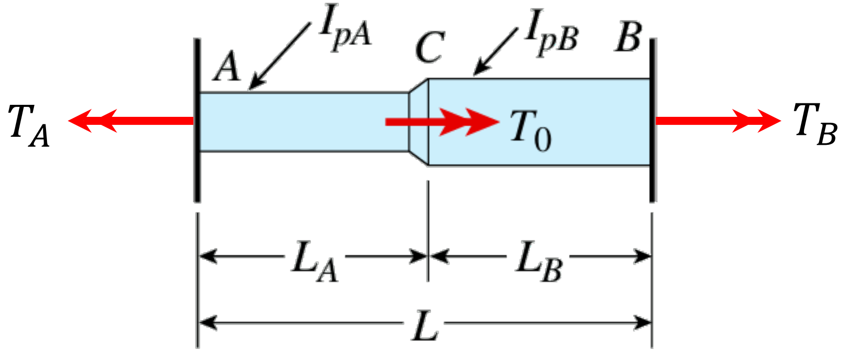


Given: T_0, d_A, d_B, L_A, L_B, I_{pA}, I_{pB}, G

Solve:

- T_A and T_B
- TMD and TDD

① FBD:



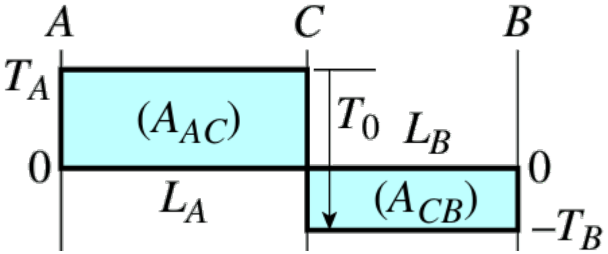
② Equilibrium equation: $-T_A + T_0 + T_B = 0$

③ Unknowns: T_A and T_B, *degree of static indeterminacy* = 1

④ Compatibility equation: $\phi_{AB} = 0$

⑤ Convert compatibility equation:

TMD:



(Jump method)

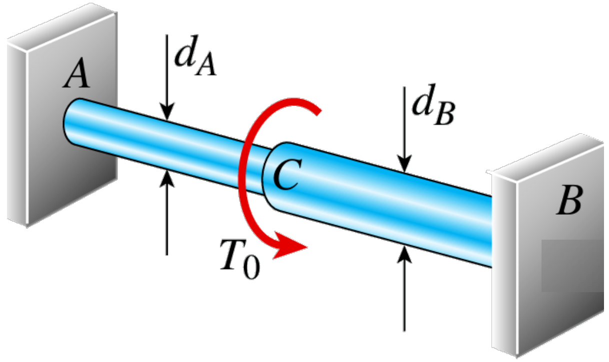
$$\phi_{AB} = \phi_{AC} + \phi_{CB} = \frac{T_A L_A}{G I_{pA}} + \frac{-T_B L_B}{G I_{pB}} = 0$$

⑥ Solved

$$T_A = T_0\left(\frac{L_B I_{pA}}{L_B I_{pA} - L_A I_{pB}}\right)$$

$$T_B = T_0\left(\frac{L_A I_{pB}}{L_B I_{pA} - L_A I_{pB}}\right)$$

Exercise

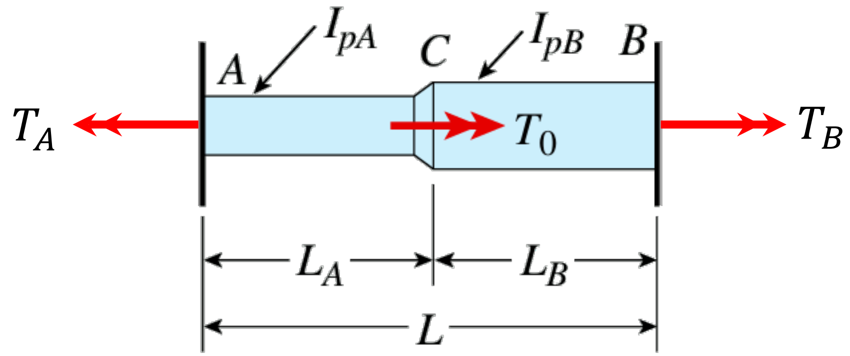


Given: $T_0, d_A, d_B, L_A, L_B, I_{pA}, I_{pB}, G$

Solve:

- T_A and T_B
- TMD and TDD

① FBD:



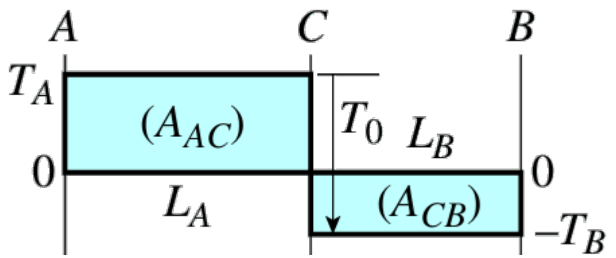
② Equilibrium equation: $-T_A + T_0 + T_B = 0$

③ Unknowns: T_A and T_B, *degree of static indeterminacy* = 1

④ Compatibility equation: $\phi_{AB} = 0$

⑤ Convert compatibility equation:

TMD:



(Jump method)

$$\phi_{AB} = \phi_{AC} + \phi_{CB} = \frac{T_A L_A}{G I_{pA}} + \frac{-T_B L_B}{G I_{pB}} = 0$$

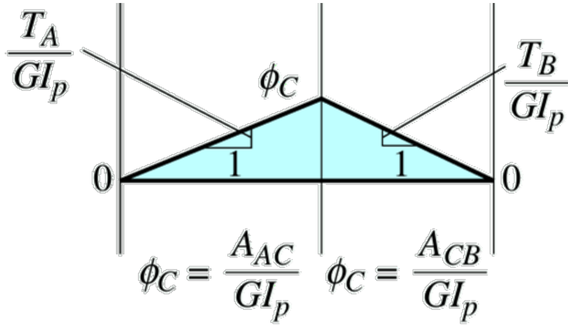
⑥ Solved

$$T_A = T_0 \left(\frac{L_B I_{pA}}{L_B I_{pA} - L_A I_{pB}} \right)$$

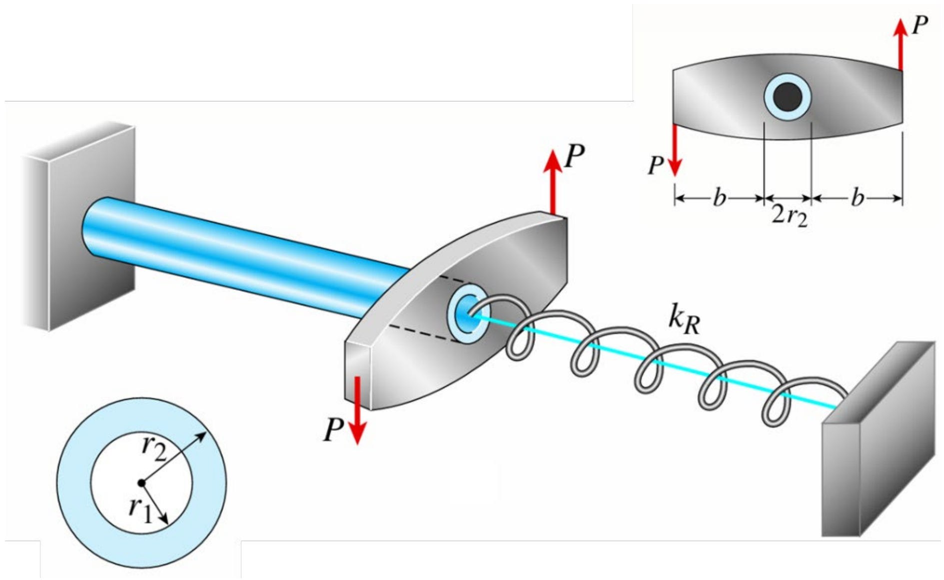
$$T_B = T_0 \left(\frac{L_A I_{pB}}{L_B I_{pA} - L_A I_{pB}} \right)$$

⑦ TDD:

$$\phi = \int_0^x \frac{T(y)}{G I_p(y)} dy$$



Hint for HW2 P6

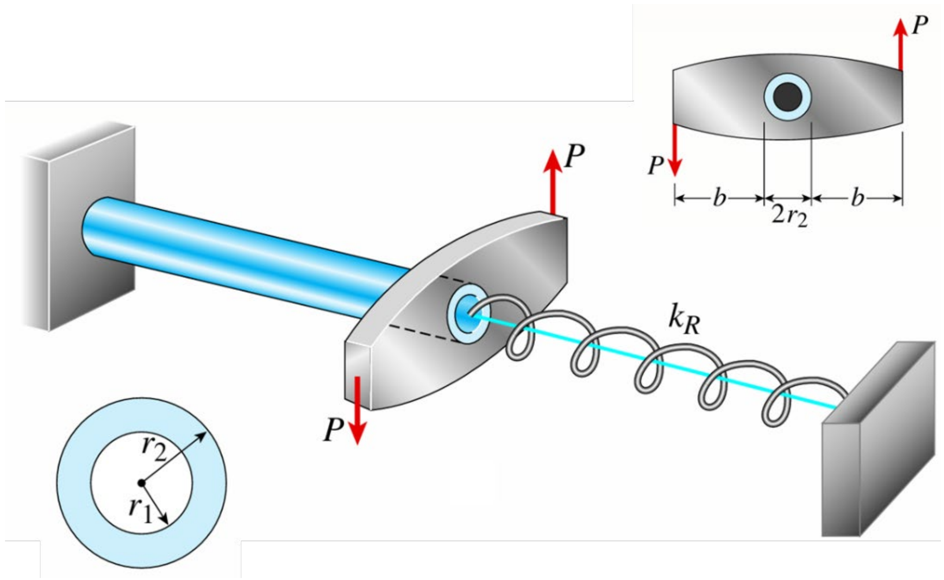


Given: P, b, r_1, r_2, G, L, k_R

Solve:

- $\tau_{\max}$

Hint for HW2 P6

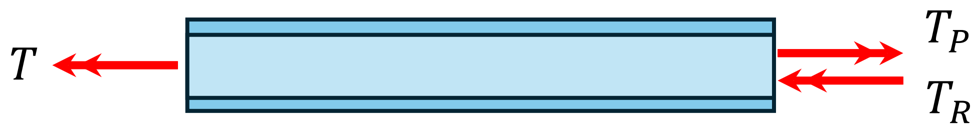


Given: P, b, r_1, r_2, G, L, k_R

Solve:

- $\tau_{\max}$

① FBD:



① Equilibrium equation: $-T + T_P - T_R = 0$

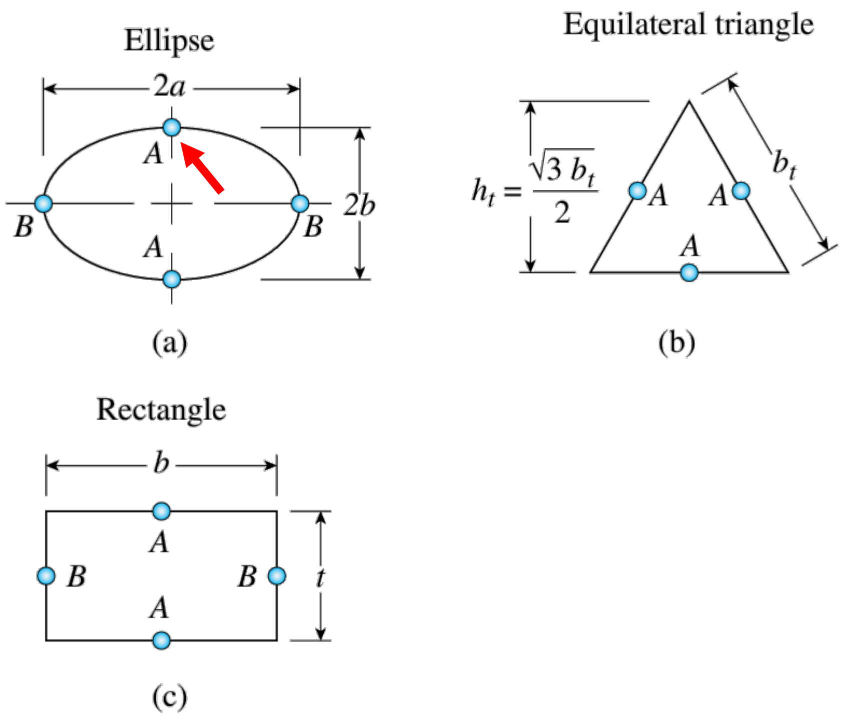
② Unknowns: T and T_R, *degree of static indeterminacy* = 1

③ Compatibility equation: $T_R = k_R|\phi|$

④ Convert compatibility equation:

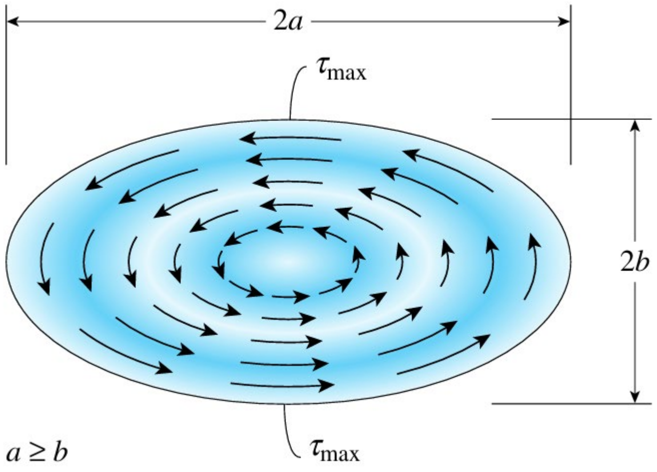
$$\phi = \frac{TL}{GI_p}, \qquad T_R = k_R\frac{TL}{GI_p}$$

*Torsion of Noncircular Prismatic Shafts



- Previous torsion formulas are valid for axisymmetric or **circular** shafts
- Planar cross-sections of noncircular shafts do not remain planar, and stress and strain distribution do not vary linearly

For elliptical cross section:



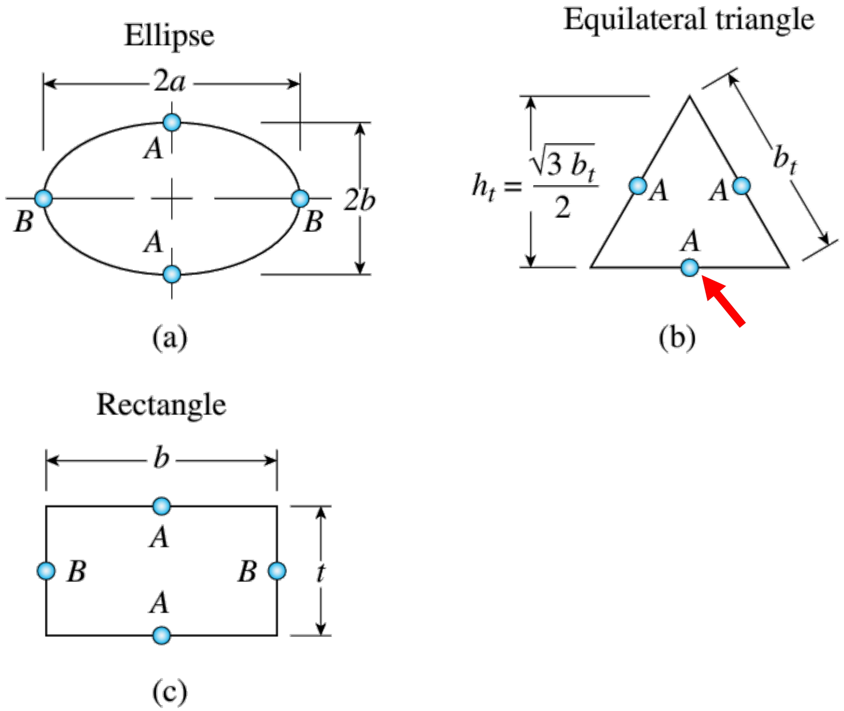
$$\tau_{max} = \frac{2T}{\pi a b^2}$$

$$\phi = \frac{TL}{GJ_e}$$

$$J_e = \frac{\pi a^3 b^3}{a^2 + b^2}$$

Torsion constant

*Torsion of Noncircular Prismatic Shafts



- Previous torsion formulas are valid for axisymmetric or **circular** shafts
- Planar cross-sections of noncircular shafts do not remain planar, and stress and strain distribution do not vary linearly

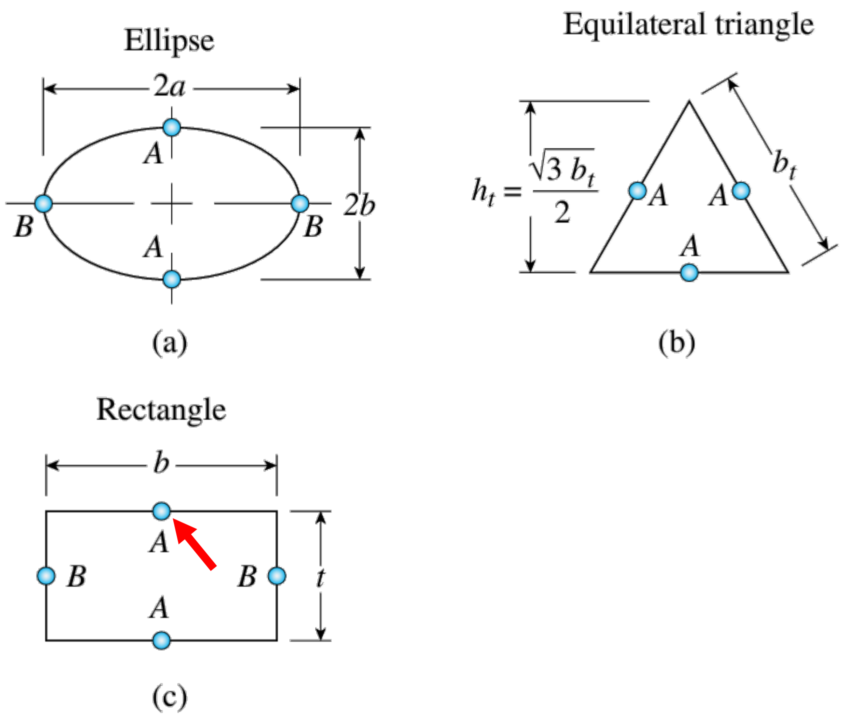
For equilateral triangular cross section:

$$\tau_{\max} = \frac{T\left(\frac{h_t}{2}\right)}{J_t} = \frac{15\sqrt{3}T}{2h_t^3}$$

$$\phi = \frac{TL}{GJ_t} = \frac{15\sqrt{3}TL}{Gh_t^4}$$

$$J_t = \frac{h_t^4}{15\sqrt{3}}$$

*Torsion of Noncircular Prismatic Shafts



- Previous torsion formulas are valid for axisymmetric or **circular** shafts
- Planar cross-sections of noncircular shafts do not remain planar, and stress and strain distribution do not vary linearly

For rectangular cross section:

$$\tau_{\max} = \frac{T}{k_1 b t^2}$$

$$\phi = \frac{TL}{GJ_r}$$

$$J_r = k_2 b t^3$$

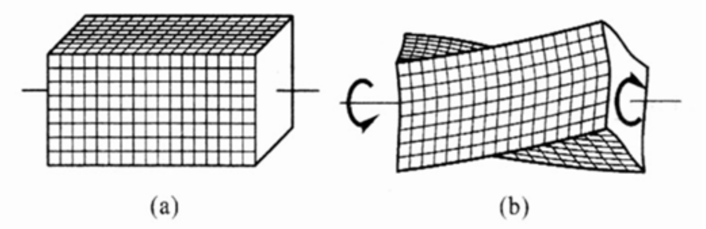


Table 3-1 Dimensionless coefficients for rectangular bars

b/t	1.00	1.50	1.75	2.00	2.50	3.00	4	6	8	10	∞
k_1	0.208	0.231	0.239	0.246	0.258	0.267	0.282	0.298	0.307	0.312	0.333
k_2	0.141	0.196	0.214	0.229	0.249	0.263	0.281	0.298	0.307	0.312	0.333